A Systematic Comparison of Large and Small Language Models in *Standalone, Hybrid & Multi-Agent* Architectures

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1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). Novel composite metric **EATS** (Efficiency-Accuracy Trade-off Score) jointly ranks architectures on quality and efficiency. **Routing** reaches 76.7% on MMLU with only 48.7% heavy-model calls; **speculative decoding** reaches 86.0%. Novel **Blackboard & Entropy Blackboard** resolve 72–93% of ARC and GSM8K queries on SLM workers alone at up to **97% accuracy** — the highest raw accuracy recorded in the study.

2 Background / Literature Review

LLMs deliver state-of-the-art reasoning but impose high latency, cost, and energy. **SLMs** are cheaper and data-sovereign, yet trail on broad reasoning tasks.

A systematic literature review of **54 collaborative SLM–LLM studies** indicates that, while hybrid and multi-agent designs are becoming more common, **cost and energy are still rarely treated as primary evaluation outcomes** alongside accuracy.

RQ1	**Performance.** How do collaborative SLM/LLM systems compare with standalone LLMs on reasoning & classification?
RQ2	**Efficiency.** How do hybrid & multi-agent designs affect cost, latency, and energy?
RQ3	**Unified metric.** Does EATS give a stable, architecture-agnostic ranking of the quality–efficiency trade-off?
RQ4	**Empirical grounding.** Do measured comparisons corroborate the trends from our 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same `Query` and returns the same `Response` carrying **accuracy, latency, tokens, cost, energy & CO_2** per query. The architecture layer is the *only* interchangeable component.

Experiment Config
architecture · benchmark · seed · models

↓

Control Plane
FastAPI · Next.js UI · runner · host-lock

↓

Architecture Layer
routing · speculative · oracle · ensemble · swarm

↓

Metrics Proxy + vLLM
latency · GPU power (NVML) · energy · CO_2 · cost

↓

Scorer → MLflow + Reports
per-item & aggregate metrics

TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	Gemma/Qwen/LLaMA 3–4B
Mid LLM	NVIDIA RTX PRO6000	96 GB	20–35B dense + MoE
Heavy LLM	NVIDIA H200	141 GB	LLaMA-70B, gpt-oss-120B

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\mathrm{EATS} = \frac{\mathit{Accuracy}}{\mathit{Accuracy} + \mathit{Penalty}}$$

$$\mathit{Penalty} = 0.5{\cdot}\mathit{Cost} + 0.3{\cdot}\mathit{Latency} + 0.2{\cdot}\mathit{Energy}$$

each ÷ standalone-LLM baseline → LLM = 1

0.5 is the theoretical ceiling for the standalone LLM.

5 Energy, Carbon & Cost

Per-request measurements via GPU power sampling (NVIDIA NVML) at the inference host.

$$\mathit{Cost} = \text{GPU rate} \cdot \text{time}$$

$$\mathit{Latency} = \mathit{\Delta t} \quad \text{per request}$$

$$E = P_{\mathrm{GPU}} \cdot \mathit{\Delta t} \quad \text{kWh}$$

$$\mathrm{CO_2} = k \cdot E$$

k = 380 gCO_2/kWh (EU avg) · E in J ÷ $3.6{\times}10^6$

6 Benchmarks & Protocol

Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking.

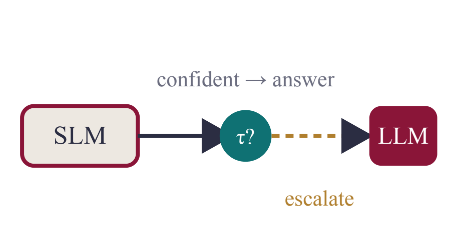
BENCHMARK	WHAT IT MEASURES
MMLU	**Broad knowledge** — 57 academic disciplines
ARC-C	**Scientific reasoning** — multi-step physical-world science
GSM8K	**Mathematical reasoning** — grade-school math, free-form
HellaSwag	**Commonsense inference** — sentence completion
TruthfulQA	**Adversarial truthfulness** — calibration vs. persuasive falsehoods

7 Architecture Taxonomy

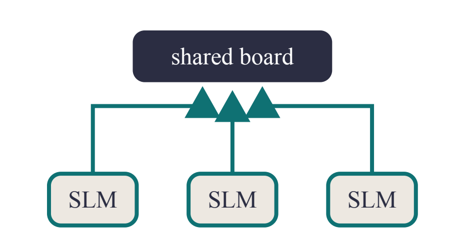
Ten architectures in three families. † marks novel project contributions introduced in the current study.

FAMILY	ARCHITECTURE	TYPE	KEY MECHANISM
STAND	Single-Model	Baseline	direct inference
STAND	MoE	Baseline	internal sparse expert activation
HYBRID	Routing	Literature	confidence SLM→LLM escalation
HYBRID	Speculative	Literature	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	Literature	proponent–opponent–arbitrator
HYBRID	Active Oracle †	Novel	targeted LLM sub-queries
HYBRID	Blackboard †	Novel	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	Novel	uncertainty-penalised bidding
MULTI	Ensemble	Literature	majority or confidence-weighted voting
MULTI	Pure Swarm †	Novel	SLM-only peer coordination; no LLM online

ROUTING

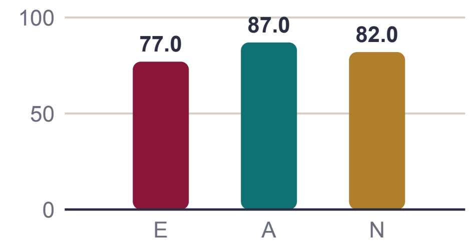


BLACKBOARD



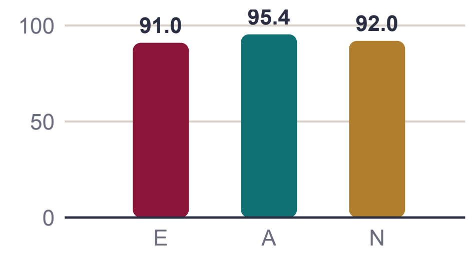
8 Benchmark Readouts

MMLU



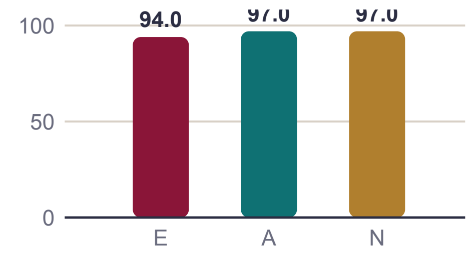
EATS Routing · 0.825
ACC gpt-oss-120b · 87.0%
NOVEL Entropy BB · 82.0% / 0.524

ARC



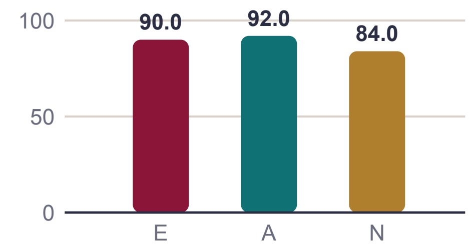
EATS Routing · 0.881
ACC gpt-oss-120b · 95.4%
NOVEL Pure Swarm · 92.0% / 0.660

GSM8K



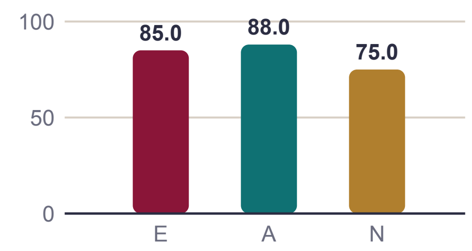
EATS Pure Swarm · 0.684
ACC Entropy BB · 97.0%
NOVEL Entropy BB · 97.0% / 0.649

HellaSwag



EATS Routing · 0.848
ACC Speculative · 92.0%
NOVEL Entropy BB · 84.0% / 0.482

TruthfulQA



EATS Routing · 0.815
ACC Speculative · 88.0%
NOVEL Entropy BB · 75.0% / 0.384

9 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	**0.82**	**0.88**	0.58	**0.85**	**0.81**
Speculative	0.81	0.85	0.51	0.83	0.78
Standalone (best MoE)	0.46	0.49	0.49	0.48	0.46
Blackboard †	0.54	0.64	0.64	0.49	0.34
Entropy Blackboard †	0.52	0.62	0.65	0.48	0.38
Active Oracle †	0.57	0.83	0.56	0.61	0.46
Debate (POA)	0.34	0.43	0.41	0.31	0.29
Pure Swarm †	0.53	0.66	**0.68**	0.50	0.48
Ensemble	0.22	0.29	0.26	0.24	0.20

Best configuration per benchmark. Darker = higher EATS · bold = benchmark winner · † novel contribution.

10 Conclusions

RQ1	Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.
RQ2	Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.
RQ3	EATS gives a consistent cross-architecture ranking: routing leads four benchmarks, and Pure Swarm leads GSM8K.
RQ4	The measurements broadly corroborate the SLR: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

SELECTED REFERENCES

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Companion SLR (54 studies, PRISMA 2020) · [6] MMLU, ARC, GSM8K, HellaSwag, TruthfulQA.

Code & experiments
github.com/DogukanTopcu